



**COST MANAGEMENT PLAN
WAH4HOSPITAL (WAH4H)**

**ASIA PACIFIC COLLEGE
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2026





This Cost Management Plan (CMP) establishes the framework, policies, and procedures by which project costs for WAH4H: Hospital Information System are planned, estimated, budgeted, monitored, controlled, and reported throughout the project lifecycle. It applies to all cost-related activities from project initiation through formal closure, covering Team Pseudoers' labor cost, infrastructure benchmarks, and all cost-related decisions requiring Project Sponsor approval.

INTRODUCTION

3.6.1.1 Document Purpose

This plan serves as the authoritative reference for all cost management decisions within the WAH4H project. It defines:

- The cost management approach, methodology, and governing principles
- Roles and responsibilities for cost control
- Cost estimation methodology and basis of estimates, including Complexity Tier calibration
- Budget structure, cost baseline, and management reserves
- Earned Value Management (EVM) metrics, thresholds, and response triggers
- Cost variance response and escalation procedures
- Cost change control process and approval authority levels
- Reporting formats and cadence

3.6.1.2 Scope of Application

This plan applies to all cost elements within the WAH4H project budget baseline, including role-based academic labor costs and infrastructure cost benchmarks. WAH assumes ₱0 liability for Team Pseudoers' simulated labor. Infrastructure costs (Proxmox deployment, Cloudflare) are WAH-managed and recurring post-closure; they are included as reference benchmarks for business model planning and investment justification as directed by Sir Robert Michael Martinez and Sir Pey Canlas.

COST MANAGEMENT APPROACH

3.6.2.1 Methodology

The WAH4H project applies a Hybrid Cost Management approach: predictive (PMBOK 6th Edition) for labor baseline establishment and academic deliverables; and adaptive (Agile/Scrum) for sprint-level effort tracking. All cost management activities are governed by PMBOK 6th Edition best practices and adapted to the academic-professional hybrid nature of the project.

3.6.2.2 Cost Management Principles

- All costs must be estimated, baselined, and formally approved before expenditure.
- The cost baseline is the approved budget used to measure project performance via Earned Value Management (EVM).



- Management Reserve (MR) is held outside the cost baseline and requires Project Sponsor approval to access.
- Contingency Reserve (CR) is included within the baseline and may be accessed by the Project Manager for identified risks.
- All cost changes exceeding defined thresholds must follow the formal Cost Change Control Process.
- Infrastructure costs are benchmark figures for WAH planning purposes; Team Pseudoers incurs no direct financial liability.

3.6.2.3 Cost Management Roles and Responsibilities

Role	Person	Cost Responsibility
Project Sponsor	Sir Pey Canlas	Approves budget baseline, management reserve access, and scope changes with cost impact > 15%
Acceptance Authority	Anna Katrina Yturalde	Reviews and approves deliverable acceptance which may trigger cost closure of work packages
Technical Supervisor	Sir Robert Michael Martinez	Approves infrastructure cost benchmarks; owns server provisioning budget decisions on WAH side
Project Manager	Elijah J. Quibin	Owns cost baseline, tracks EVM metrics, approves variances within threshold, escalates above threshold
Lead Developer / DevOps	Jhon Lloyd Nicolas	Provides labor effort estimates for development work packages; reports progress against plan
QA Engineer	Mariyah Chavez	Provides labor estimates for QA, infrastructure validation, and UAT work packages
Scrum Master / Agile Developer	John Kenneth L. Jajurie	Reports sprint velocity; provides frontend effort estimates; tracks sprint-level cost performance

COST ESTIMATION METHODOLOGY

3.6.3.1 Estimation Approach

Labor cost estimates are derived using a Bottom-Up Estimation approach: effort hours are estimated at the work package level per WBS (v1.1), then aggregated by team member and sprint. Work package effort is calibrated using a Complexity Tier system (Low / Normal / High / Critical) that correlates tier to planned hour range and Acceptance Criteria count, ensuring



estimation consistency across all 111 work packages. Infrastructure cost benchmarks use Analogous Estimation based on Akamai Linode pricing as the cost ceiling reference, as confirmed by Sir Robert Michael Martinez.

Tier	Hour Range	AC Count	When Applied	Example Work Packages
Low	0.5–2.0h	1–2 AC	Single-purpose deliverable. Companion document. Revision with minimal changes. No independent research. Assignee is domain expert.	Agile Team Values, Sprint ceremony outputs, minor revisions
Normal	2.1–5.0h	3–4 AC	Multi-section document with moderate research. First draft with some coordination. References other documents. Assignee has working familiarity.	Comms Plan, Scope Plan, Risk Plan, Requirements Plan, most planning documents
High	5.1–8.0h	4–5 AC	Complex multi-condition deliverable. Diagram or visual creation required. Strong cross-document dependencies. External stakeholder input required. Significant research or technical depth.	Project Charter, WBS, Cost Plan, SRS, System Design Document, PBL2 modules, Feasibility Study
Critical	8.1h+	5+ AC or unplanned event	Unplanned scope-defining event with cascading impact across baselines. Full team involvement required. External client or sponsor dependency.	CR-001 Scope Confirmation (10.0h — 3 members involved)

3.6.3.2 Labor Rate Basis

All team members are billed at role-based market-reference rates derived from Philippine salary benchmarks (JobStreet PH, Indeed PH, PayScale PH 2026). These rates replace the earlier uniform ₱500/hour placeholder and reflect the distinct skill profiles of each team member. WAH assumes ₱0 labor cost liability per project charter constraints.

Each rate is derived using the standard monthly-to-hourly conversion (monthly benchmark ÷ 160 billable hours) and calibrated to the seniority level appropriate for healthcare IT project work. Elijah Josh R. Quibin (Project Manager) is benchmarked at the upper bound of JobStreet PH's IT Project Manager range as of April 2026 (₱95,000/month ÷ 160 = ₱594/hr), reflecting full



ownership of all project management deliverables, stakeholder coordination, and cost control. Jhon Lloyd Nicolas (Lead Developer / DevOps) is benchmarked at the lead-level DevOps rate from NodeFlair PH's 2025 Salary Report ($\text{₱}100,000/\text{month} \div 160 = \text{₱}625/\text{hr}$), appropriate for his ownership of all 11 module builds, Docker/Proxmox infrastructure, and FHIR R4 backend architecture. Mariyah Vanna Monique C. Chavez (QA Engineer) is benchmarked at the QA/Testing median from NodeFlair PH as of February 2026 ($\text{₱}58,000/\text{month} \div 160 = \text{₱}363/\text{hr}$), suitable for UAT oversight, compliance validation, and documentation quality review. John Kenneth L. Jajurie's original rate of $\text{₱}95/\text{hr}$ was identified as significantly below the market average for a Scrum Master / Agile Developer role in the Philippine IT sector; the revised rate of $\text{₱}309/\text{hr}$ is benchmarked using PayScale PH 2026 data for a Certified Scrum Master with Agile Software Development skills ($\text{₱}593,100/\text{year} \div 12 = \text{₱}49,425/\text{month} \div 160$), reflecting his dual responsibility for sprint facilitation and frontend UI development.

Role	Team Member	Hourly Rate	Source	Notes
Project Manager	Elijah Quibin	₱594 / hr	JobStreet PH (April 2026)	PM-owned and Elijah split tasks
Lead Dev / DevOps	Jhon Lloyd Nicolas	₱625 / hr	NodeFlair PH 2025 Salary Report	Development and DevOps tasks
QA Engineer	Mariyah Chavez	₱363 / hr	JobStreet PH Feb 2026	QA, UAT, and validation tasks
Scrum Master / Agile Developer	John Kenneth L. Jajurie	₱309 / hr	PayScale PH 2026 (CSM + Agile Dev skills)	Scrum ceremonies and frontend support

3.6.3.3 Work Package Effort Distribution — Planned Hours per Person per Sprint

Total planned project labor is 344 hours across 111 work packages and 3 sprints. Distribution by sprint and team member is shown below, derived from the WBS work package register (WAH4H_WP_Hours_v2_2.xlsx).

Sprint	Elijah (hrs)	Jhon (hrs)	Mariyah (hrs)	Jajurie (hrs)	Sprint Total
Sprint 1 (Mar 23 – Apr 22)	21.25	18.5	14.5	14.25	68.5
Sprint 2 (Apr 23 – May 5)	10.25	11.0	7.0	3.75	32.0



Sprint 3 (May 6 – May 27)	44.25	119.0	51.5	28.75	243.5
TOTAL	75.75	148.5	73.0	46.75	344.0

Note: Jhon Lloyd Nicolas carries 119.0 planned hours in Sprint 3 — the highest individual sprint allocation — reflecting his primary ownership of all 11 module development and infrastructure work packages. Planned hours are strictly derived from the work package register and do not represent financial commitments by WAH.

PROJECT BUDGET

3.6.4.1 Cost Baseline

The cost baseline is the approved, time-phased budget against which project cost performance is measured. It excludes Management Reserve. Budget is computed using role-based rates applied to planned hours per member.

Component	Amount (PHP)	Notes
Elijah — 75.75 hrs × ₱594/hr	₱44,995.50	Project Manager
Jhon — 148.5 hrs × ₱625/hr	₱92,812.50	Lead Developer / DevOps
Mariyah — 73.0 hrs × ₱363/hr	₱26,499.00	QA Engineer
Jajurie — 46.75 hrs × ₱309/hr	₱14,445.75	Scrum Master / Agile Developer
Labor Cost (One-Time)	₱185,158.50	Sum of all member planned labor
Contingency Reserve (30% of Labor)	₱55,547.55	PM-controlled; for identified risks
COST BASELINE (Labor + CR)	₱240,706.05	Approved spending authority (excl. MR)
Management Reserve (5% of Baseline)	₱12,035.30	Sponsor-controlled; for unknown unknowns
TOTAL PROJECT BUDGET (Labor)	₱252,741.35	Maximum authorized labor budget
Infrastructure — Annual (WAH-Managed)	₱135,192.00	\$164/mo × 12 × ₱56.50 + ISP; WAH bears post-closure
REFERENCE TOTAL (Labor Budget + 1-Year Infrastructure)	₱236,731.80	Reference figure for WAH business model planning only

3.6.4.2 Recurring Infrastructure Cost Benchmark (WAH-Managed)

Akamai Linode pricing is used as the cost ceiling reference for cloud infrastructure planning. Proxmox is the confirmed on-premise deployment platform. All recurring infrastructure costs are



WAH-managed and WAH-funded post-closure. Included per Sir Pey Canlas's directive for investment justification and revenue modeling.

Item	Specification	USD/Mo	PHP/Mo	PHP/Year
Akamai Linode Server (Ref.)	320GB Storage, 8 CPUs, 16GB RAM	\$144.00	₱8,136.00	₱97,632.00
Cloudflare (Tunnel + DDoS)	Existing service	\$20.00	₱1,130.00	₱13,560.00
Redundant ISP (Secondary Line)	Mid-tier broadband/fiber for failover	—	₱2,000.00	₱24,000.00
TOTAL RECURRING (Cloud + ISP)		\$164.00	₱11,266.00	₱135,192.00

Redundant ISP cost is a Philippine market estimate (e.g., PLDT Fiber/Globe Fiber mid-tier ~₱2,000/month). Actual cost to be confirmed by Sir Robert Michael Martinez. Conversion rate: \$1 ≈ ₱56.50 (May 2026).

3.6.4.3 One-Time Hardware Capital Expenditure (Proxmox Setup)

The following hardware items are required for a production-grade Proxmox on-premise deployment. These are one-time capital expenditures (CapEx) borne entirely by WAH. Team Pseudoers bears no liability for WAH-owned infrastructure costs. Costs are Philippine market estimates as of May 2026; Sir Robert Michael Martinez is responsible for procurement.

Hardware Item	Purpose / Justification	Est. Cost (PHP)	Owner	P#
Primary Proxmox Server	Main host for WAH4H VMs. ≥16GB RAM, 320GB+ storage, 8 CPUs.	₱80,000 – ₱150,000	Sir Robert Michael Martinez	P1
UPS — Primary Server	Protects against brownouts and power interruptions. Min 1000VA/600W; recommended 2000VA.	₱8,000 – ₱25,000	Sir Robert Michael Martinez	P1
Redundant ISP Hardware (Dual-WAN Router)	Router/modem for secondary ISP. Enables automatic failover.	₱3,000 – ₱8,000	Sir Robert Michael Martinez	P1
High-Speed Managed Gigabit Switch	Min 8-port gigabit managed switch. Enables VLAN segmentation.	₱3,000 – ₱12,000	Sir Robert Michael Martinez	P1



Secondary / Backup Physical Server	Dedicated backup server for high-availability. ≥8GB RAM, 1TB+ storage.	₱40,000 – ₱100,000	Sir Robert Michael Martinez	P2
UPS — Backup Server	Dedicated UPS for backup server and network switch.	₱5,000 – ₱15,000	Sir Robert Michael Martinez	P2

CapEx Summary	Low Estimate (PHP)	High Estimate (PHP)
Primary Proxmox Server	₱80,000	₱150,000
UPS — Primary Server	₱8,000	₱25,000
Redundant ISP Hardware (Dual-WAN Router)	₱3,000	₱8,000
High-Speed Managed Gigabit Switch	₱3,000	₱12,000
Secondary Backup Server	₱40,000	₱100,000
UPS — Backup Server	₱5,000	₱15,000
TOTAL ONE-TIME HARDWARE CAPEX	₱139,000	₱310,000

P1 = required at initial deployment. P2 = strongly recommended; may be phased in post-initial deployment. All CapEx is WAH-borne — not included in Team Pseudoers' project cost baseline.

3.6.4.4 Total Infrastructure Cost Summary (WAH Reference)

Component	Type	Low Est. (PHP)	High Est. (PHP)
One-Time Hardware CapEx (Section 3.6.4.3)	One-Time	₱139,000	₱310,000
Annual Recurring OpEx — Year 1 (Section 3.6.4.2)	Annual	₱135,192	₱135,192
TOTAL YEAR 1 (CapEx + OpEx)	Combined	₱274,192	₱445,192
Annual Recurring OpEx — Year 2 onwards	Annual	₱135,192+	₱135,192+
Team Pseudoers Labor Contribution	One-Time (Academic)	₱0 (WAH liability)	₱0 (WAH liability)

3.6.4.5 Contingency Reserve Rate Justification

The 30% Contingency Reserve (₱55,547.55) is supported by four converging sources:

#	Source	Justification
1	PMI Estimating Guidance	For software projects with high technical complexity, schedule compression, and external dependency risk, reserve analysis supports contingency reserves in the 20–35% range. WAH4H exhibits all three



		conditions: FHIR R4 compliance complexity, a fixed June 2026 deadline, and unconfirmed Proxmox server access at the time of estimation.
2	AACE International RP No. 40R-08	For Class 4–5 estimates with low-definition scope and early-stage planning, AACE cites expected accuracy ranges of –15% to +50%, supporting reserves above 25%. WAH4H’s scope involved unconfirmed server specifications and pending interoperability requirements at the time of estimation.
3	EHR Implementation Benchmarks	Healthcare IT implementations involving EHR-adjacent systems consistently report cost overruns of 20–40% above baseline due to integration complexity, regulatory compliance rework, and stakeholder-dependency delays — all of which apply directly to WAH4H’s PhilHealth eClaims integration and FHIR R4 alignment requirements.
4	Defense Panel Recommendation	During the AY 2025–2026 T3 Finals Defense, the project adviser recommended a contingency rate of 30–40% based on direct experience with comparable healthcare IT projects. The team adopted 30% as the conservative lower bound of that range, consistent with all three references above.

The 30% rate supersedes the earlier 10% placeholder and is the controlling contingency figure for all cost baseline and EVM calculations in this document.

3.6.4.6 Contingency Reserve — Scenario Coverage

The 30% Contingency Reserve (P\$55,547.55) is sized to cover identified cost risk scenarios. Hardware failure at or immediately after formal turnover was specifically raised as a scenario the project budget must be prepared for. The reserve covers: emergency server repair or temporary cloud redeploy (linked to R-01 and R-22 in the Risk Register); post-UAT defect rework exceeding planned effort (R-12); late scope changes requiring additional development hours (R-04); and integration complexity overruns in FHIR R4 compliance and eClaims implementation. See §3.8 Risk Register for full risk entries. See §8.1.8 for the hardware recovery procedure included in the Technical Manual handover.

MEASURING PROJECT COSTS — EARNED VALUE MANAGEMENT

3.6.5.1 EVM Framework Overview

The WAH4H project employs Earned Value Management (EVM) per PMBOK 6th Edition to objectively measure cost and schedule performance. EVM data is collected at the end of each sprint and at major milestone gates. Measurement is based on work package completion percentage as assessed by the PM and validated by the Lead Developer. Actual Cost is computed using role-based rates (Table 135B) applied to actual hours logged per work package.



3.6.5.2 EVM Key Variables

Metric	Full Name	Definition	Formula / Source
PV	Planned Value	Budgeted cost of scheduled work at any given point in time	Time-phased cost baseline
EV	Earned Value	Budgeted cost of work actually performed (completed)	% complete × BAC per WBS
AC	Actual Cost	Actual labor effort incurred, computed by role-based rate	Sprint actual hrs × member role rate
BAC	Budget at Completion	Total planned cost for all project work (cost baseline)	₱240,706.05 (excl. MR)
ETC	Estimate to Complete	Remaining expected cost to finish the project	(BAC – EV) / CPI or direct re-estimate
EAC	Estimate at Completion	Forecast of total project cost at completion	AC + ETC
VAC	Variance at Completion	Difference between BAC and EAC	BAC – EAC

3.6.5.3 EVM Performance Indices

Index	Full Name	Formula	Ideal Value	Interpretation
CPI	Cost Performance Index	EV / AC	≥ 1.0	<1 = over budget; >1 = under budget
SPI	Schedule Performance Index	EV / PV	≥ 1.0	<1 = behind schedule; >1 = ahead
CV	Cost Variance	EV – AC	0 (positive is good)	Negative = cost overrun
SV	Schedule Variance	EV – PV	0 (positive is good)	Negative = schedule slip
TCPI	To-Complete PI	(BAC – EV) / (BAC – AC)	≤ 1.10	>1.10 signals target unrealistic; triggers re-baseline discussion



3.6.5.4 Performance Thresholds & Response Triggers

Zone	CPI Range	SPI Range	Status	Required Action
GREEN (On Track)	≥ 1.00	≥ 1.00	Within tolerance	Document in sprint report; continue as planned
YELLOW (Watch)	0.90–0.99	0.85–0.99	Minor variance	PM logs variance; identifies root cause; presents recovery plan at next sprint review
ORANGE (Warning)	0.80–0.89	0.75–0.84	Significant variance	PM initiates corrective action within 48 hrs; brief Sir Robert Michael Martinez; re-baseline sub-tasks if needed; document in risk register
RED (Critical)	< 0.80	< 0.75	Critical overrun	Immediate escalation to Sir Pey Canlas; formal variance analysis; change request required; management reserve access requires Sponsor approval

3.6.5.5 Cost Variance Acceptance Criteria

Criteria	Threshold	Basis / Rationale
Acceptable Cost Variance (%)	$\pm 10\%$ of planned cost per sprint	Standard tolerance for academic-scale project; within normal estimation uncertainty
Acceptable Cost Variance (₱)	$\pm \text{₱}55,547.55$ cumulative (10% of $\text{₱}185,158.50$ baseline)	Aligned to 10% contingency reserve size; exceedance triggers CR drawdown
Single Work Package Overrun	$\leq 20\%$ of WP planned budget	PM discretion to approve; escalate to Sponsor if $> 20\%$
Contingency Reserve Trigger	CR accessed when $CV < -\text{₱}2,000$ on any phase	PM authorization required; logged in Cost Change Log
Management Reserve Trigger	MR accessed when $CPI < 0.80$ or $EAC > BAC + MR$	Sponsor authorization required via formal change request
EAC Tolerance (At Completion)	$EAC \leq \text{₱}252,741.35$ (BAC + MR)	Any forecast exceeding total project budget requires Sponsor re-baseline approval
TCPI Ceiling	$TCPI \leq 1.10$	Above 1.10 signals remaining budget target is unrealistic; triggers re-baseline discussion



REPORTING FORMAT

3.6.6.1 Cost Reporting Cadence

Report	Frequency	Audience	Prepared By	Medium
Sprint Cost Report	Per sprint (end of sprint)	Team Pseudoers	PM	Internal Scrum artifact
EVM Status Update	Bi-weekly	PM + Sir Robert Michael Martinez	PM	Email / Messenger
Milestone Cost Summary	At each milestone gate (M1–M11)	Sir Pey Canlas, Sir Robert Michael Martinez, Ms. Anna Katrina Yturalde	PM	Email + formal doc attachment
Cost Variance Report	On threshold breach (YELLOW/ORANGE/RED)	Sir Robert Michael Martinez / Sir Pey Canlas	PM	Email with formal variance analysis
Final Cost Report	At project closure	All stakeholders	PM	Included in Project Closeout Report

3.6.6.2 Sprint Cost Report Contents

- Sprint number and reporting period
- Planned Value (PV), Earned Value (EV), Actual Cost (AC) for the sprint — computed using role-based rates
- Cost Variance (CV) and Schedule Variance (SV)
- CPI and SPI — current sprint and cumulative
- EAC, ETC, and VAC projections
- Contingency Reserve balance remaining
- Any cost variances, root cause, and corrective actions taken
- Risk register updates related to cost

COST VARIANCE RESPONSE PROCESS

When a cost variance is detected (CPI, SPI, or CV outside GREEN thresholds), the following response process must be followed in sequence:

Step	Action	Detail	Owner
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1	Detect	EVM metrics computed at sprint end or milestone gate reveal CPI/SPI outside GREEN zone	PM
2	Classify	Assign to YELLOW / ORANGE / RED zone per thresholds in Section 3.6.5.4	PM
3	Analyze	Root cause analysis: scope creep, underestimation, dependency delay, rework, or external factor	PM + Lead Developer
4	Document	Log variance in Sprint Cost Report and Risk Register with root cause, impact, and proposed response	PM
5	Respond	Execute corrective action per zone: YELLOW = recovery plan; ORANGE = re-baseline sub-tasks + notify Sir Robert Michael Martinez; RED = escalate to Sir Pey Canlas + formal change request	PM (+ Sponsor if RED)
6	Monitor	Track recovery metrics for 1 sprint; if CPI/SPI not returning to GREEN after 2 sprints, escalate one zone higher	PM
7	Close	Document resolution in Sprint Report; update Risk Register; confirm reserve balances	PM

COST CHANGE CONTROL PROCESS

3.6.8.1 Principles

- No unplanned cost may be incurred without following the change control process.
- All changes impacting the cost baseline must be formally documented, assessed for impact, and approved before implementation.
- Verbal approvals are not sufficient; all cost change approvals must be documented in writing (email or signed change request).
- The PM maintains the Cost Change Log as part of the project records.

3.6.8.2 Approval Authority Matrix

Change Type	Threshold	Approver	Turnaround
Effort reallocation within same WBS	No net cost change	PM	Immediate
Minor cost overrun — within Contingency Reserve	CV up to — ₱55,547.55	PM	Within 1 sprint
Contingency Reserve drawdown	Any CR access	PM (must log)	Within sprint cycle
Scope change with cost impact	Any positive cost delta	PM → Sir Pey Canlas (Sponsor)	48–72 hrs



Management Reserve access	$CPI < 0.80$ or $EAC > BAC$	PM submits formal CR → Sir Pey Canlas	72 hrs max
Re-baseline (full budget revision)	$EAC > BAC + MR (> ₱252,741.35)$	Sir Pey Canlas formal approval	Sponsor decision meeting
Infrastructure cost change (WAH-side)	Any	Sir Robert Michael Martinez	WAH internal decision

3.6.8.3 Change Request Process Flow

- Step 1 — Initiate: PM or team member identifies a cost impact event and documents it in the Cost Change Request form (cost impact, justification, WBS reference).
- Step 2 — Assess: PM conducts impact analysis (effect on cost baseline, CR/MR balance, EAC, schedule impact) and classifies per approval authority matrix.
- Step 3 — Review: Appropriate approver reviews the request within the defined turnaround window.
- Step 4 — Decide: Approved, rejected, or deferred with written documentation.
- Step 5 — Implement: If approved, PM updates cost baseline (or CR/MR balance), Sprint Backlog, and EVM data.
- Step 6 — Log: Change logged in Cost Change Log with ID, date, description, impact, decision, and approver signature.

BUDGET REVISION, INFLATION & FOREIGN EXCHANGE ADJUSTMENTS

This section governs the procedures and triggers for revising WAH4H infrastructure cost benchmarks in response to economic factors including Philippine Peso/US Dollar exchange rate fluctuations and general inflation. All financial risk arising from these factors is borne exclusively by the WAH organization. WAH assumes ₱0 liability for Team Pseudoers' simulated labor, while WAH owns all post-closure infrastructure costs under any cost revision scenario — this is an absolute constraint established in the Project Charter and is non-negotiable.

3.6.9.1 Financial Liability Disclaimer

Team Pseudoers (BSCS-SS231) assumes zero (₱0) financial liability for any and all project costs, including but not limited to: labor compensation, infrastructure subscription fees, exchange rate losses, inflation-driven cost increases, or any other monetary obligation arising from the WAH4H project. All infrastructure costs are WAH-managed and WAH-funded. The cost figures in this document are academic simulations and operational benchmarks only. Authority for all budget revisions, cost increases, and financial decisions related to infrastructure rests solely with WAH (Sir Pey Canlas, Sir Robert Michael Martinez).

3.6.9.2 Baseline Exchange Rate & Inflation Reference

Parameter	Baseline Value	Source / Notes
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USD/PHP Exchange Rate	\$1 ~ ₱56.50	BSP approximate rate, May 2026
Linode Server (Cloud Ref.)	\$144.00 / month	Akamai Linode pricing, May 2026. Cost ceiling per Sir Robert Michael Martinez.
Cloudflare (Tunnel + DDoS)	\$20.00 / month	Existing WAH service. Fixed recurring cost.
Redundant ISP (Secondary Line)	₱2,000.00 / month	PH market estimate. Confirmed by Sir Robert Michael Martinez at deployment.
Total Recurring OpEx Baseline	₱11,266/mo = ₱135,192/yr	Cloud (\$164/mo) + Redundant ISP (₱2,000/mo) at \$1=₱56.50
PHP CPI Inflation Reference	~3.0–4.5% annual (BSP target band)	Applied to PHP-denominated recurring costs for multi-year projections

3.6.9.3 Revision Trigger Thresholds

Zone	Trigger Type	Threshold	Severity	Required Action
GREEN	FX Rate	±3% from ₱56.50 (₱54.78–₱58.23)	Monitor	Note in monthly report; no action unless sustained >30 days
YELLOW	FX Rate	±5% from baseline (₱53.68–₱59.33)	Watch	PM issues formal FX Advisory to Sir Robert Michael Martinez with revised cost projections
ORANGE	FX Rate	±10% from baseline (₱50.85–₱62.15)	Warning	PM issues Cost Revision Notice to Sir Pey Canlas + Sir Robert Michael Martinez
RED	FX Rate	>+15% from baseline (<₱48.03 or >₱65.00)	Critical	Immediate advisory to Sir Pey Canlas; WAH initiates formal infrastructure budget re-evaluation
YELLOW	Annual Inflation (PHP)	Infrastructure PHP cost increases >5% YoY	Watch	PM includes inflation-adjusted projections in Final Cost Report; WAH reviews budget
ORANGE	Annual Inflation (PHP)	Infrastructure PHP cost increases >10% YoY	Warning	Formal Cost Revision Notice to Sir Pey Canlas; WAH reviews contract terms with providers
YELLOW	Linode Price Change	Linode increases server tier price >5%	Watch	PM advises Sir Robert Michael Martinez; re-evaluate tier or alternative providers



ORANGE	Linode Price Change	Linode increases server tier price >15%	Warning	Formal advisory to Sir Pey Canlas + Sir Robert Michael Martinez; evaluate migration to alternative provider
GREEN	Cloudflare Price Change	Cloudflare increases <\$5/month	Monitor	Note in cost report; no action required
YELLOW	Cloudflare Price Change	Cloudflare increases >\$5/month or plan restructure	Watch	Advise Sir Robert Michael Martinez; evaluate plan alternatives

3.6.9.4 Budget Revision Procedure

Step	Action	Detail	Owner
1	Detect & Verify	Monitor BSP FX rates monthly (last Friday of each month) and Linode/Cloudflare pricing at each subscription cycle.	PM
2	Calculate Impact	Compute revised monthly and annual infrastructure cost at new rate. Quantify delta in PHP vs. ₱135,192/year baseline.	PM
3	Classify Zone	Apply thresholds from Section 3.6.9.3 to determine GREEN/YELLOW/ORANGE/RED zone.	PM
4	Prepare Advisory	Draft Cost Revision Advisory: trigger type, current vs. baseline rate, revised annual cost projection, delta, and recommended WAH action options.	PM
5	Communicate	YELLOW: email Sir Robert Michael Martinez. ORANGE: email Sir Robert Michael Martinez + Sir Pey Canlas. RED: immediate escalation to Sir Pey Canlas with urgent flag.	PM
6	WAH Decision	Sir Robert Michael Martinez and/or Sir Pey Canlas evaluate options: absorb cost, downgrade tier, switch provider, or migrate fully to Proxmox on-premise.	Sir Robert Michael Martinez / Sir Pey Canlas
7	Update Records	PM updates infrastructure cost benchmark table in this CMP (via Change Request), logs event in Cost Change Log, and reflects revised projections in Final Cost Report.	PM



8	Confirm Liability	Regardless of outcome, reconfirm in writing that Team Pseudoers bears ₱0 liability for Team Pseudoers' simulated labor; WAH owns all post-closure infrastructure costs for any cost increase.	PM + Sir Pey Canlas
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3.6.9.5 Multi-Year Recurring Infrastructure Cost Projections (Reference)

Scenario	FX Rate	USD/Mo (Cloud)	PHP/Mo (Cloud+ISP)	Annual PHP	vs. Yr 1
Year 1 — Baseline	₱56.50	\$164.00	₱11,266	₱135,192	—
Year 2 — Inflation +4%	₱56.50 (stable)	\$164.00	₱11,717	₱140,604	+₱5,412 (+4.0%)
Year 3 — Inflation +4% YoY	₱56.50 (stable)	\$164.00	₱12,186	₱146,232	+₱11,040 (+8.2%)
Year 2 — FX ₱60.00	₱60.00 (+6.2%)	\$164.00	₱11,840	₱142,080	+₱6,888 (+5.1%)
Year 2 — FX ₱65.00	₱65.00 (+15.0%)	\$164.00	₱12,660	₱151,920	+₱16,728 (+12.4%)
Year 2 — Full Proxmox Only	N/A (no USD)	\$0.00	₱2,000 (ISP only)	₱24,000	-₱111,192 (-82%)

Full Proxmox on-premise eliminates all USD-denominated costs post-hardware-investment, leaving only the secondary ISP line as recurring cost. This is the lowest-risk scenario for long-term FX exposure.

3.6.9.6 Hardware CapEx Revision Triggers

Hardware Item	Revision Trigger	Impact	Decision Authority
UPS (Primary or Backup)	Brownout frequency higher than expected, or UPS fails within warranty/post-warranty period	Upgrade to higher VA capacity; ₱5,000–₱25,000 additional	Sir Robert Michael Martinez
Redundant ISP Hardware	Primary ISP changes infrastructure or dual-WAN router becomes incompatible	Router replacement; ₱3,000–₱15,000	Sir Robert Michael Martinez



High-Speed Network Switch	Hospital site expands; more devices need LAN ports or higher-tier managed switch features required	Switch upgrade; ₱5,000–₱20,000	Sir Robert Michael Martinez
Backup Physical Server	Primary server failure rate higher than tolerated, or WAH expands to additional hospital sites	Additional or replacement server; ₱40,000–₱150,000; may trigger re-scope of Proxmox cluster	Sir Robert Michael Martinez + Sir Pey Canlas
All Hardware	PHP general hardware price inflation >10% YoY affects procurement cost for replacements	Adjust WAH hardware reserve budget; prioritize P1 items first	Sir Pey Canlas

3.6.9.7 Cost Mitigation Options for WAH

Option	Description	Cost Impact	Recommended When
Full Proxmox On-Premise	Run WAH4H entirely on WAH-owned Proxmox hardware; no cloud subscription	Eliminates \$144/mo Linode; retains \$20/mo Cloudflare = \$240/yr	FX rate exceeds RED threshold (>₱65.00) or Linode price >15%
Downgrade Linode Tier	Switch from Dedicated 16GB to Shared 8GB plan (~\$60/mo)	Reduces server cost by ~58% (~\$84/mo savings)	Load testing confirms lower specs sufficient for demo environment
Alternative Provider	Migrate to DigitalOcean, Vultr, or Hetzner for comparable specs at lower cost	Hetzner EU: ~\$40–60/mo for similar specs; requires migration effort	Linode price >10% or sustained FX depreciation >10%
Absorb & Budget	Accept increased cost as-is and adjust WAH operating budget	No infrastructure change required	FX <10% and inflation <5% YoY; cost increase is manageable

GLOSSARY OF TERMS

Term	Definition
AC	Actual Cost — total labor cost incurred for completed work, computed as actual hours × member role-based rate



BAC	Budget at Completion — total planned cost of the project (₱240,706.05: role-based labor ₱185,158.50 + 30% Contingency Reserve ₱55,547.55), excluding Management Reserve
BSP	Bangko Sentral ng Pilipinas — Philippine central bank; official source for USD/PHP exchange rate data
Budget Revision	Formal update to infrastructure cost benchmark figures in this CMP, approved via Change Control Process; does not affect the academic labor cost baseline
CMP	Cost Management Plan — this document; governs all cost planning, measurement, and control activities
Complexity Tier	Work package classification (Low / Normal / High / Critical) used to calibrate planned hours and AC count at estimation. Defined in Section 3.6.3.1.
CPI	Cost Performance Index — EV/AC ; measures cost efficiency (≥ 1.0 = on or under budget)
CR	Contingency Reserve — ₱55,547.55 PM-controlled budget for identified risks (within cost baseline)
CV	Cost Variance — $EV - AC$; positive = under budget, negative = over budget
EAC	Estimate at Completion — forecasted total project cost
ETC	Estimate to Complete — expected remaining cost to finish the project
EV	Earned Value — budgeted cost of work actually performed ($\% \text{ complete} \times \text{BAC}$ per WBS)
EVM	Earned Value Management — integrated cost/schedule performance measurement framework (PMBOK 6th Edition)
MR	Management Reserve — ₱12,035.30 Sponsor-controlled contingency for unknown unknowns (outside baseline)
PV	Planned Value — time-phased budgeted cost of scheduled work
Role-Based Rate	Member-specific hourly rate reflecting Philippine market benchmarks: Elijah ₱594/hr (PM), Jhon ₱625/hr (Lead Dev), Mariyah ₱363/hr (QA), Jajurie ₱309/hr (SM)
SPI	Schedule Performance Index — EV/PV ; measures schedule efficiency (≥ 1.0 = on or ahead of schedule)
SV	Schedule Variance — $EV - PV$; positive = ahead of schedule, negative = behind
TCPI	To-Complete Performance Index — efficiency required to complete within remaining budget; > 1.10 signals unrealistic target
VAC	Variance at Completion — $BAC - EAC$; positive = under budget at completion, negative = overrun forecast
WBS	Work Breakdown Structure — hierarchical decomposition of all project deliverables and work packages (WAH4H WBS v1.1)



APPROVAL & AUTHORIZATION

Approval/signature collection for this section is consolidated in Section 9, Approvals and Authorization. The content, acceptance criteria, and review requirements in this section remain valid and are reviewed as part of the full master document approval.

Role	Name	Signature	Date
Project Sponsor	Sir Pey Canlas		
Technical Supervisor	Sir Robert Michael Martinez		
Acceptance Authority	Ms. Anna Katrina Yturralde		
Project Manager	Elijah Josh R. Quibin		

Sponsor Acceptance

Approved by the Project Sponsor:

Sir Pey Canlas
Executive Director, Wireless Access for Health (WAH)

Date: _____

